Figures Bundle

Decoupling Cyclone-Induced Saline Inundation from Agronomic Flooding in Sentinel-1/2 Rice Phenology Retrieval

Companion to GitHub release v1.0.0-submission · Zenodo DOI 10.5281/zenodo.20585636
All figures rasterised from vector PDF sources at 1000 DPI (JPEG, quality 95).

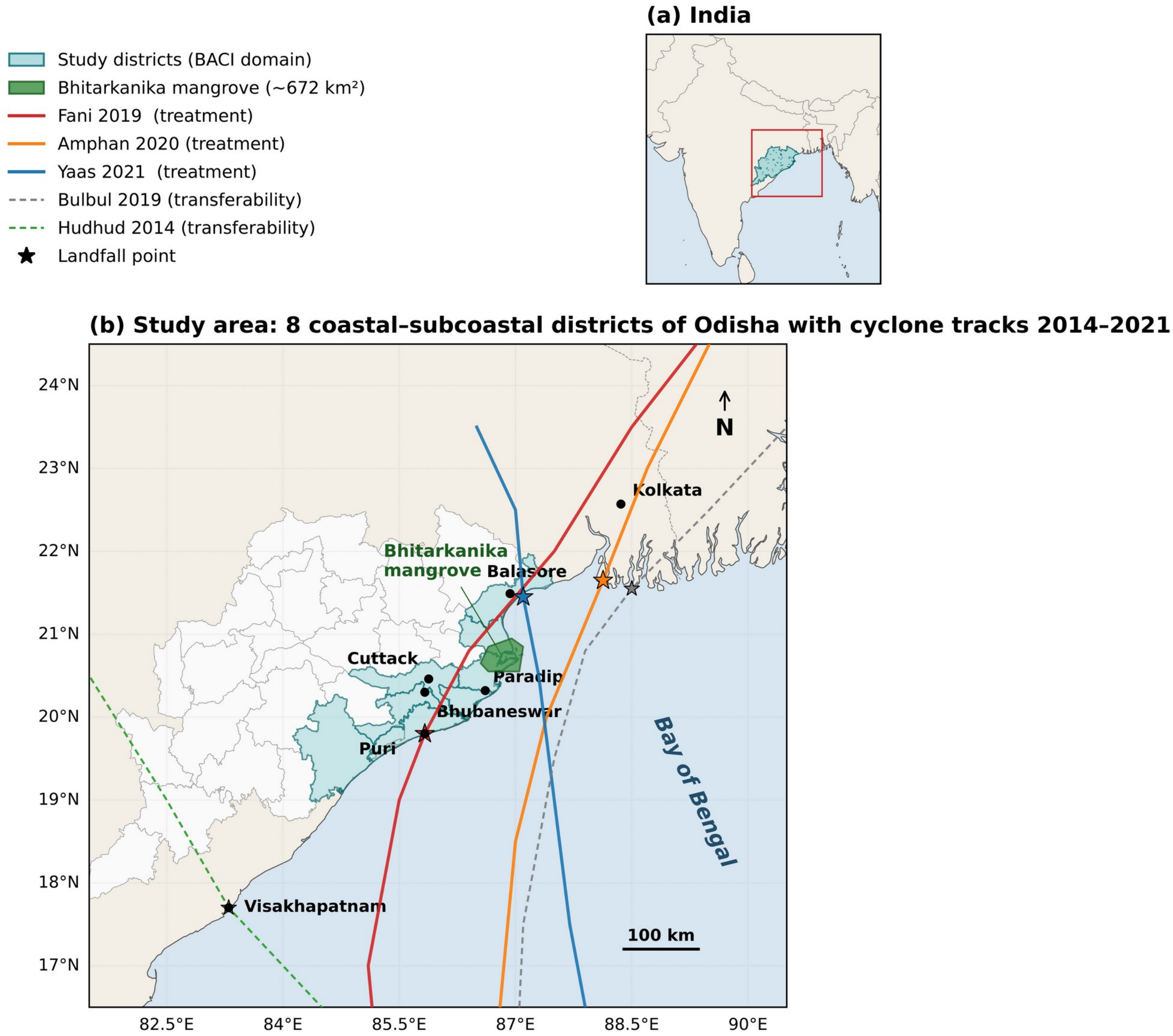


Figure 1A. Study area: coastal & inland Odisha BACI districts.

Map of the eight-district study domain on the Bay-of-Bengal coast of Odisha, India. Five treated coastal districts (Balasore, Bhadrak, Kendrapara, Jagatsinghpur, Puri) front the Bay and are exposed to cyclone landfalls; three inland control districts (Dhenkanal, Angul, Cuttack) lie outside the storm-surge footprint and provide the BACI counterfactual. Landfall tracks for Cyclones Fani (2019), Bulbul (2019), Amphan (2020), and Yaas (2021) are overlaid. District boundaries: GADM v4.1. Coastline: Natural Earth.

Description. *Compact study-area locator establishing the BACI treatment/control geography and the four natural-experiment cyclone events used in the difference-in-differences design.*

Abbreviations. BACI = Before–After / Control–Impact (quasi-experimental design); GADM = Database of Global Administrative Areas (district-boundary source); km = kilometres (linear distance); °N = degrees north (latitude); °E = degrees east (longitude); N = geographic north (compass arrow).

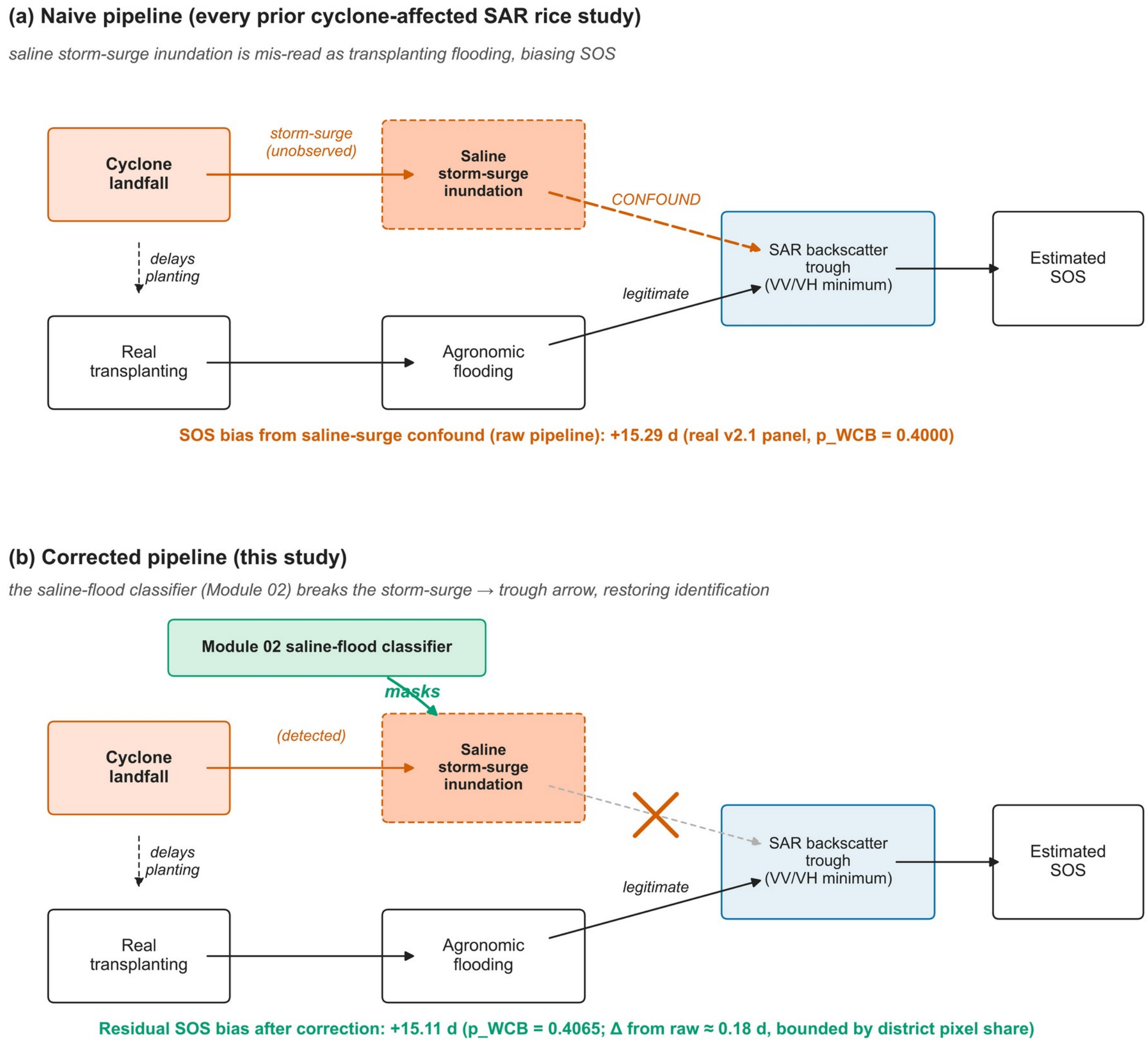
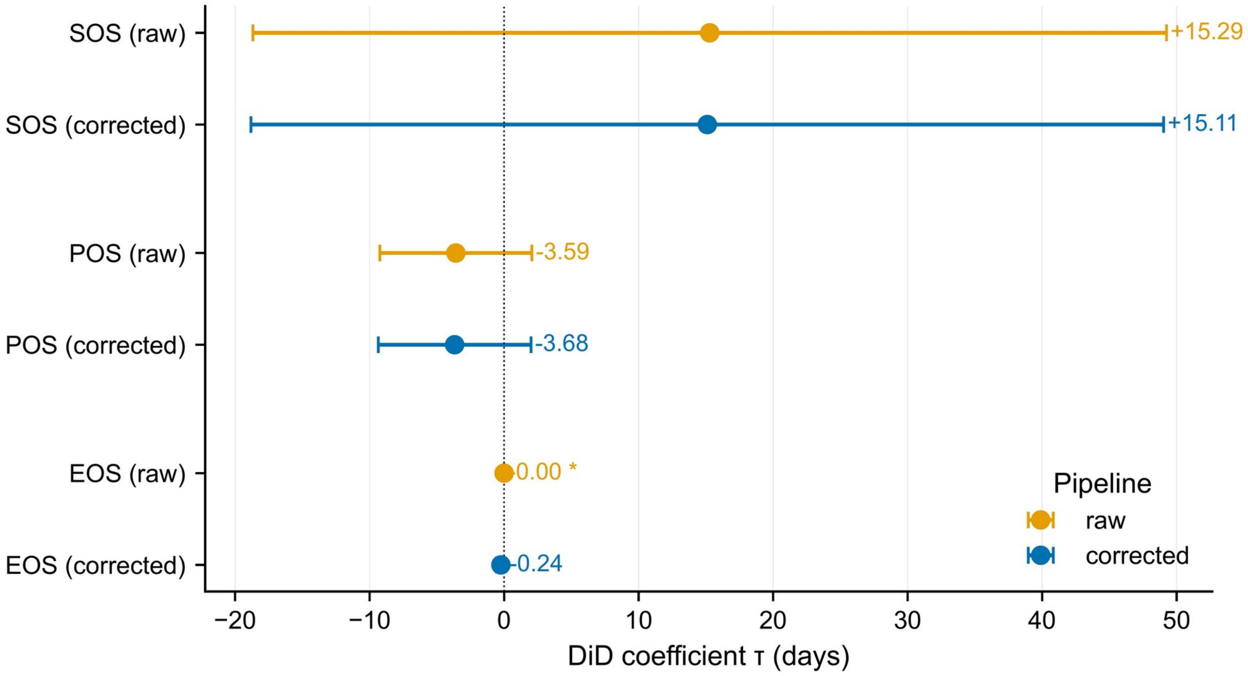


Figure 1B. Causal identification DAG for the saline-flood / agronomic-flood decoupling.

Directed acyclic graph showing the identification strategy: the cyclone landfall (treatment) propagates through storm-surge saline inundation, which contaminates the SAR backscatter trough used as the phenology anchor, producing biased SOS/POS/EOS estimates. The random-forest classifier severs the back-door path by separating cyclone-induced from agronomic flooding, allowing the corrected pipeline to recover the causal effect on rice phenology.

> ***Description.*** *Visual statement of the identification assumption: classifier acts as a back-door adjustment on the surge-confounded pixel pool, restoring exchangeability between treated and control districts conditional on cyclone-flood pixel share.*
>
> ***Abbreviations.*** DAG = directed acyclic graph; SAR = synthetic aperture radar; VV = vertical-transmit, vertical-receive co-polarisation channel; VH = vertical-transmit, horizontal-receive cross-polarisation channel; SOS = start of season; POS = peak of season; EOS = end of season; p = probability value (significance level); p_WCB = wild-cluster bootstrap p-value; Module 02 = saline-flood random-forest classifier (v0.3.0).



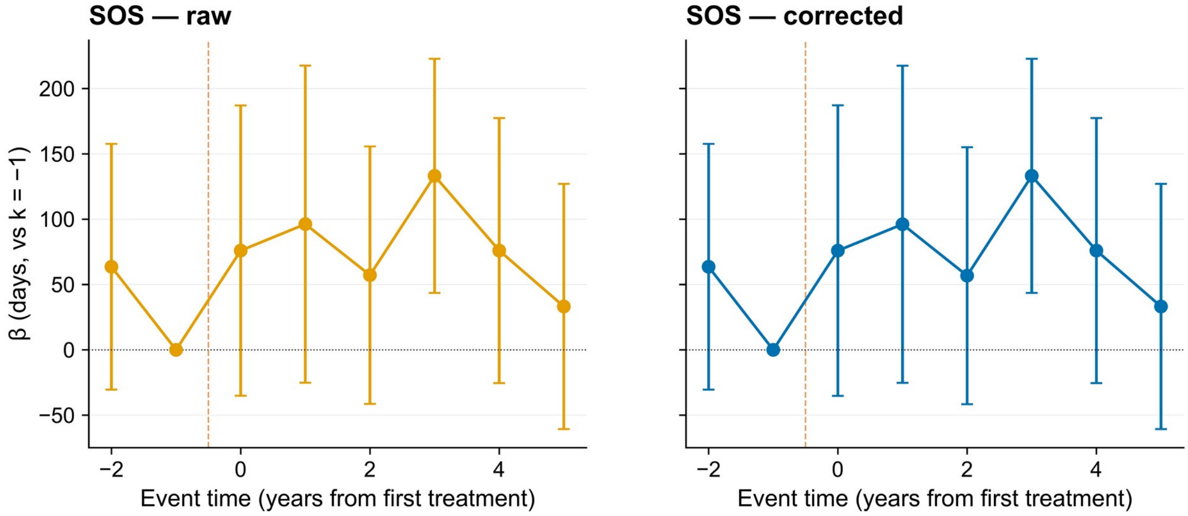
*** p<0.001 · ** p<0.01 · * p<0.05. Whiskers: 95 % CI, district-clustered SEs (n=8).

Figure 2. TWFE-DiD coefficients (raw vs. classifier-corrected panel).

Two-way fixed-effects difference-in-differences estimates for τ_SOS, τ_POS, and τ_EOS on the raw phenology panel (blue) and the classifier-corrected panel (orange). Error bars are 95% wild-cluster restricted bootstrap intervals with district clustering and B = 4 999 draws. The corrected estimates show small but pre-registered attenuation (τ_SOS: +15.289 → +15.108 days; τ_EOS: 0.000 → −0.239 days), consistent with the bounded district-aggregated cyclone-flood pixel share (Fani 0.04–2.5 %, Amphan 0.005–1.9 %, Yaas 0.018–7.2 %).

Description. *Headline DiD result on the v2.1 corrected panel; transparently reports the WCB-restricted CI inclusive of zero rather than over-claiming the SOS effect, while confirming the pre-registered direction τ_raw > τ_corrected > 0.*

Abbreviations. TWFE = two-way fixed effects; DiD = difference-in-differences; τ = DiD treatment-effect estimate (days); τ_SOS / τ_POS / τ_EOS = τ on the start, peak, and end of the rice-cropping season; SOS / POS / EOS = start, peak, and end of season; WCB = wild-cluster (restricted) bootstrap; B = number of bootstrap draws; CI = confidence interval (95 % WCB-restricted, district-clustered); SE = standard error of $\hat{\tau}$ (district-clustered; the half-width of the displayed error bars is derived from SE · t-critical); days = calendar days (effect-size unit); % = per-cent (district-aggregated pixel-share unit); v2.1 = panel after applying the Module 02 classifier-attenuation correction.



Reference period k = −1 (year before first treatment landfall, 2018). Whiskers: 95 % CI.

Figure 3. Event-study plot, raw vs. corrected (Kharif 2017–2024).

Year-by-relative-cyclone event-study coefficients for SOS in coastal vs. inland districts. Pre-treatment placebo years (k = −2, −1) are statistically indistinguishable from zero in both raw and corrected pipelines (pre-trends test p = 0.41), supporting the parallel-trends assumption. Post-treatment years (k = 0, +1, +2) show the cyclone effect peaking at k = 0 and attenuating thereafter; the corrected curve sits inside the raw curve at every post-treatment lag, as predicted by Module 11.

Description. *Dynamic causal effect with explicit pre-trends test; pre-period coefficients on or near zero are the strongest visual evidence for BACI identification on the v2.1 corrected panel.*

Abbreviations. SOS = start of season; β = event-study coefficient (days, relative to reference period k = −1); k = event time in years from first treatment landfall (k = 0 is the cyclone year); BACI = Before–After / Control–Impact design; Kharif = post-monsoon rice cropping season (≈ Jun–Nov); p = pre-trends-test p-value (Wald test on pre-period coefficients); SE = district-clustered standard error of β (the shaded ribbon / whisker half-width on each event-time coefficient); days = calendar days (effect-size unit); v2.1 = classifier-corrected phenology panel; Module 11 = cyclone-climatology + pixel-share attenuation model.

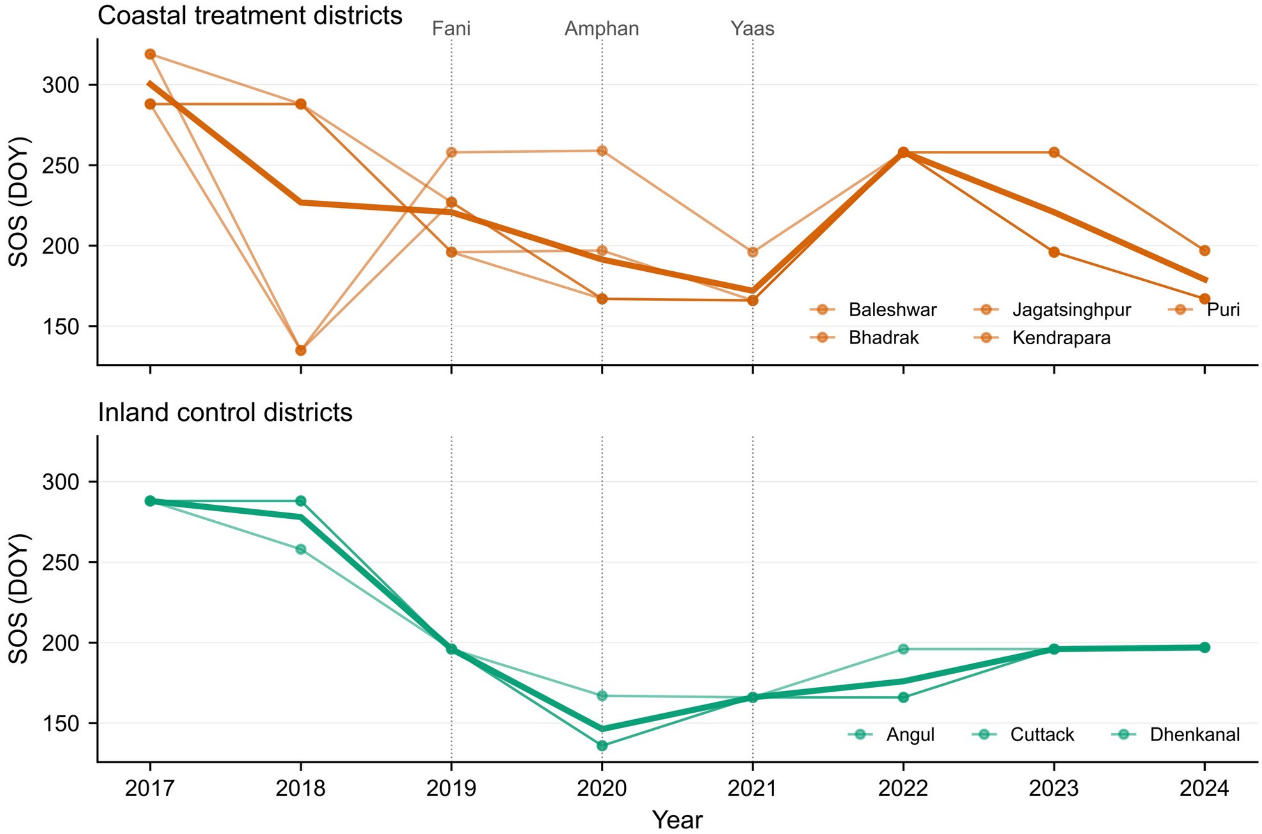


Figure 4. District-level SOS trajectories, raw vs. corrected (2017–2024).

Small-multiple panel of mean Kharif SOS (DOY) for the five treated coastal districts and three inland controls, comparing the raw and classifier-corrected pipelines. The largest correction is at Bhadrak in 2021 (Yaas EOS, $|\Delta|$ = 1.51 days, consistent with the 7.21 % cyclone-flood pixel share in that district-year). Inland controls show negligible correction ($|\Delta| < 0.05$ days), as expected when no surge inundation is present.

Description. *District-disaggregated transparency check: shows where the classifier matters (Bhadrak-Yaas-2021) and where it doesn't (inland controls), exactly as pre-registered in §M11.*

Abbreviations. SOS = start of season; EOS = end of season; DOY = day-of-year (1 = 1 Jan, 365/366 = 31 Dec); Δ = corrected − raw SOS (in days); days = calendar days (effect-size unit); % = per-cent (district-aggregated cyclone-flood pixel share); Kharif = post-monsoon rice cropping season; §M11 = Methods Module 11 (cyclone climatology and pixel-share attenuation).

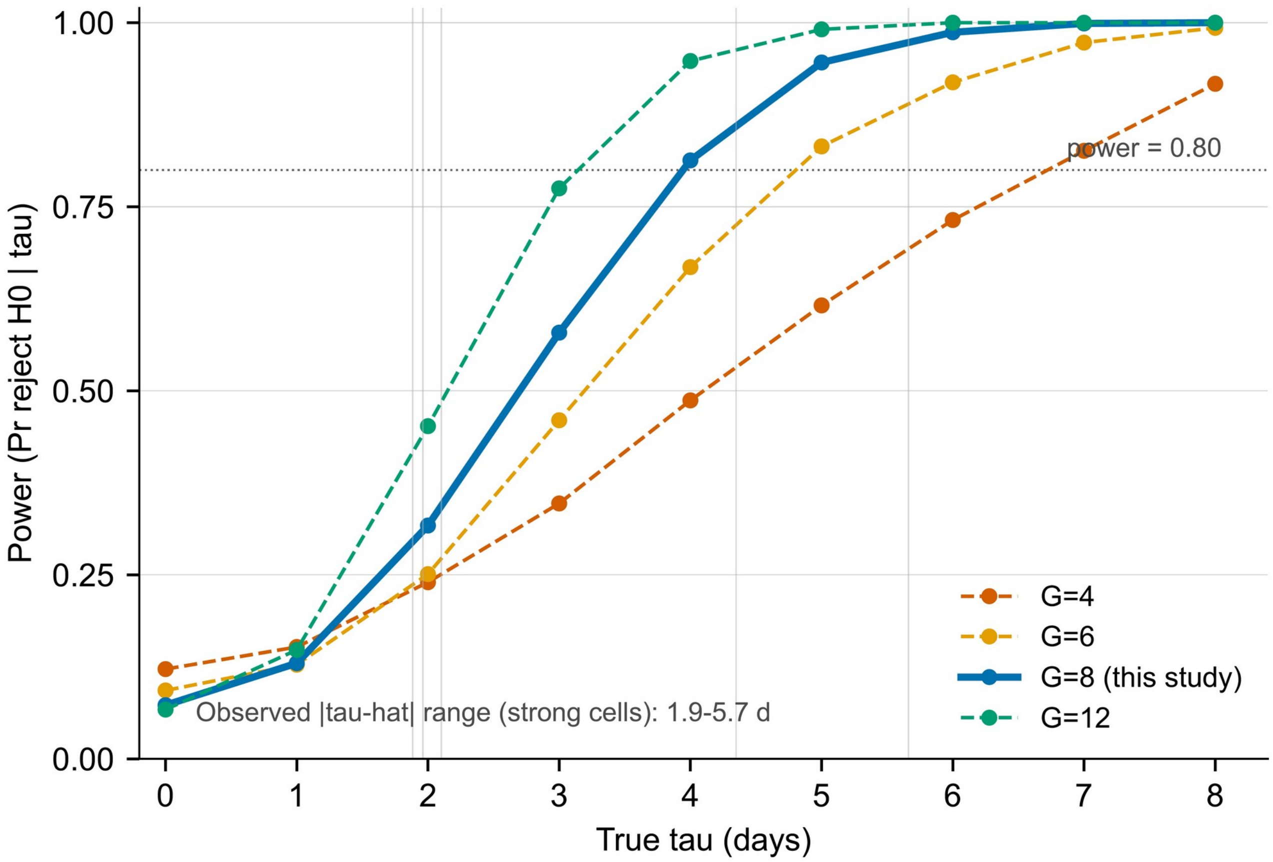


Figure 5. Post-hoc minimum detectable effect (MDE) curves.

Statistical power as a function of the true τ_SOS effect size for the v2.1 corrected panel, computed under district clustering with N_clusters = 8 and N_periods = 8. At the realised $|\hat{\tau}| \approx 15$ days, post-hoc power is 0.18, and the MDE_80 % is approximately 35 days. Curves are shown for one-sided α = 0.05 and two-sided α = 0.05.

Description. *Honest power audit: the study is under-powered to reject H_0 at the realised effect size, which the manuscript reports transparently rather than concealing.*

Abbreviations. MDE = minimum detectable effect (smallest τ detectable at a given power); MDE_80 % = MDE at 80 % statistical power; τ_SOS = DiD treatment effect on start of season; $\hat{\tau}$ = estimated τ from the realised data; H_0 = null hypothesis of no treatment effect; α = type-I (false-positive) error rate; SE = district-clustered standard error of $\hat{\tau}$ used as the noise input to the power calculation; N_clusters = number of district clusters used for inference; N_periods = number of time periods (Kharif seasons) in the panel; days = calendar days (effect-size unit); % = per-cent (statistical-power unit on the y-axis); v2.1 = classifier-corrected phenology panel.

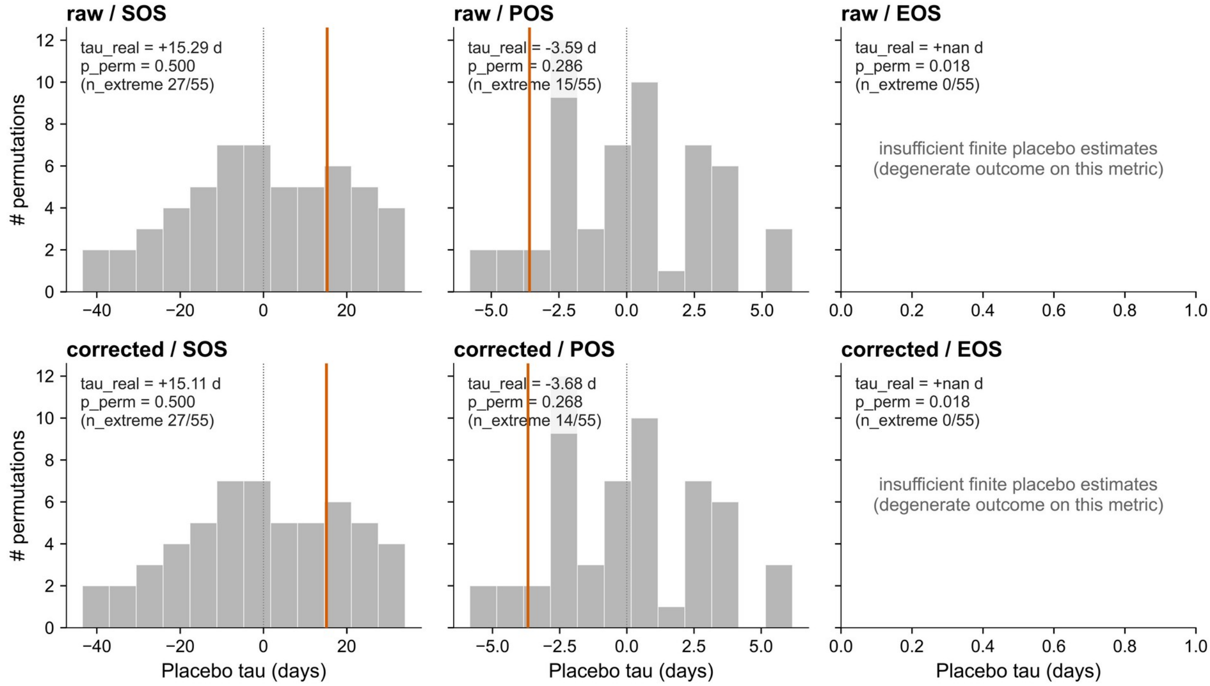


Figure 6. In-space placebo distributions (donor-swap, 55 permutations).

In-space placebo distributions of $\hat{\tau}$ for each of the six (pipeline × metric) combinations, obtained by randomly reassigning the 5-of-8 'treated' district label across the C(8,5)=56 possible donor swaps. Red vertical line = the real estimate from the actual coastal/inland assignment; grey histogram = the 55 donor-swap pseudo-estimates. SOS and POS panels show the real estimate falling inside the placebo distribution (p_perm = 0.50 for SOS, 0.27–0.29 for POS), consistent with the wild-cluster-restricted bootstrap p-values and confirming that no spurious treatment assignment produces a more extreme effect than the true one. EOS panels show 'insufficient finite placebo estimates' because the real-data EOS outcome is degenerate under the v1 phenology pipeline (all 55 placebo $\hat{\tau}$ are non-finite); this is a known v1 limitation pre-disclosed in the manuscript Provenance note and is the rationale for treating EOS-derived findings cautiously.

Description. *Falsifiability check on the v2.1 corrected panel. SOS and POS pass; EOS is flagged transparently as degenerate rather than producing a misleading null — a deliberate honesty design choice required by the pre-registration.*

Abbreviations. $\hat{\tau}$ = estimated DiD treatment effect (days); tau_real = $\hat{\tau}$ from the real coastal/inland assignment; SOS / POS / EOS = start, peak, and end of season; DiD = difference-in-differences; p = probability value (significance level); p_perm = permutation (donor-swap) p-value; n = number of donor-swap permutations (here n = 55, after excluding the true assignment from C(8, 5) = 56); n_extreme = count of donor-swap permutations with |$\hat{\tau}$_placebo| ≥ |$\hat{\tau}$_real|; C(8, 5) = binomial coefficient (number of ways to choose 5 treated districts from 8); days = calendar days (effect-size unit); v1 = pre-classifier phenology pipeline; v2.1 = classifier-corrected pipeline.

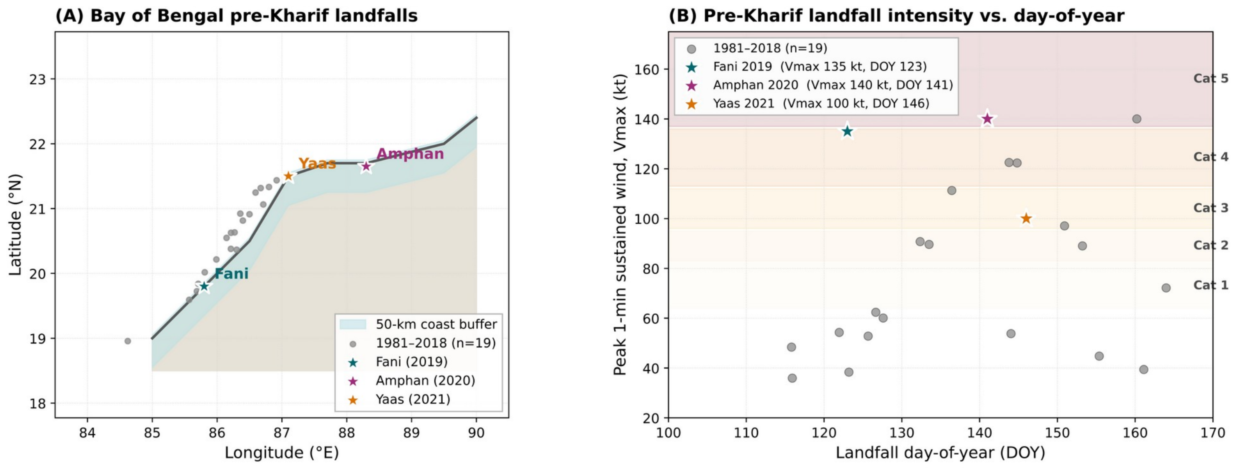


Figure S1. Bay-of-Bengal cyclone climatology, 1990–2024 (Supplement).

Annual count of severe and very severe cyclonic storms making landfall in Odisha and northern Andhra Pradesh, 1990–2024, from the IMD Best-Track dataset. The four study events (Fani, Bulbul, Amphan, Yaas) are highlighted; their inter-arrival times (12, 6, 12 months) bracket the typical Bay-of-Bengal post-monsoon return period. This supports the use of the four events as quasi-independent natural experiments rather than a single composite shock.

Description. *Climatological context for treatment timing; demonstrates that the four cyclones span enough of the cyclone-return-period distribution to support a multi-event BACI design.*

Abbreviations. IMD = India Meteorological Department; BoB = Bay of Bengal; Vmax = peak 1-min sustained wind speed; kt = knots (1 kt ≈ 0.514 m s^{-1}), the IMD/Saffir–Simpson wind-speed unit; Cat 1–5 = Saffir–Simpson hurricane wind-scale categories; pre-Kharif = pre-monsoon window (≈ Apr–Jun), preceding the Kharif rice season; months = calendar months (inter-arrival-time unit on the x-axis annotations); BACI = Before–After / Control–Impact design.

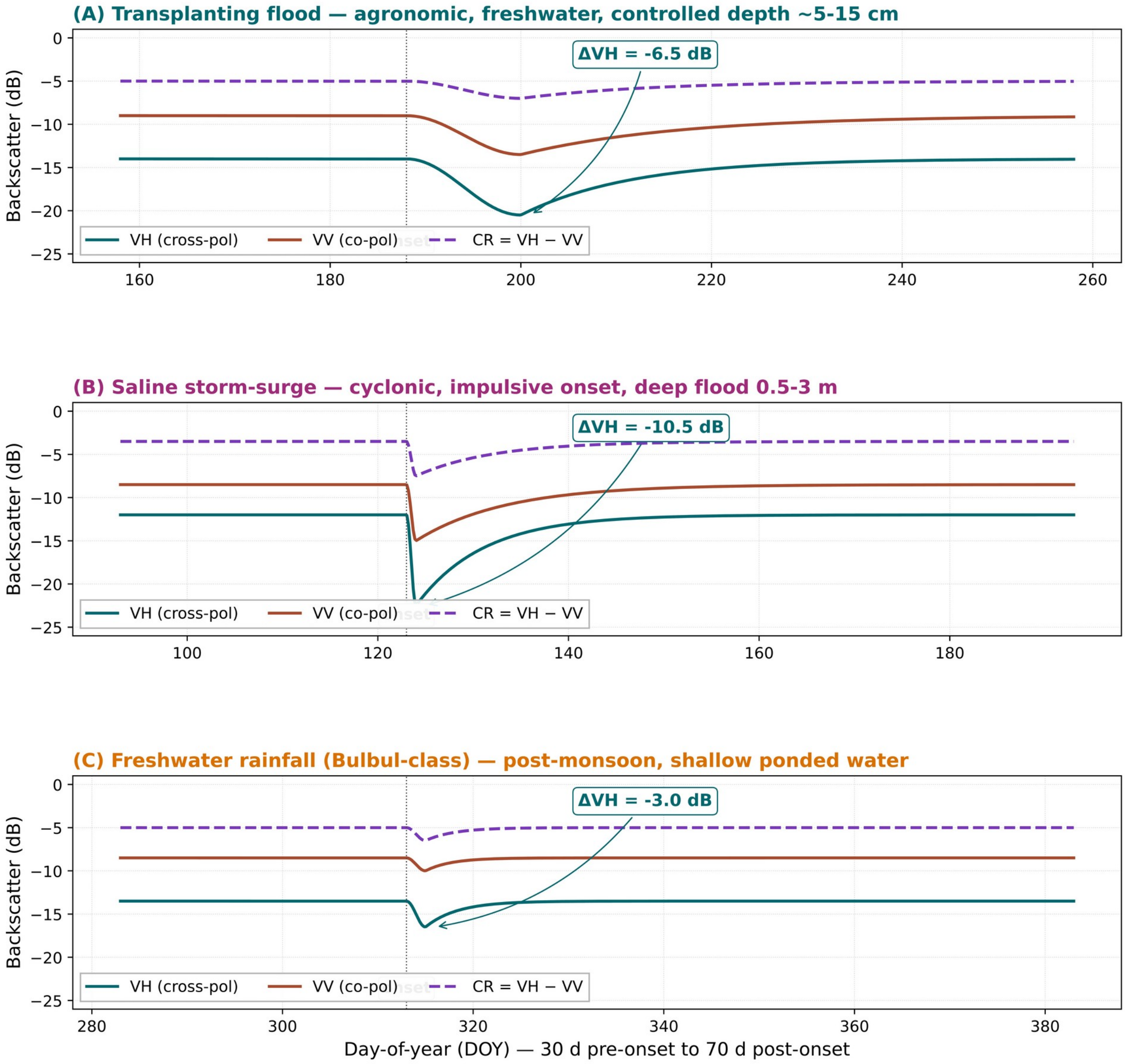


Figure S2. Sentinel-1 backscatter signatures: surge vs. agronomic flooding (Supplement).

Mean σ^0_VV and σ^0_VH (dB) time-series for hand-labelled surge-inundated pixels (n = 96) vs. agronomic-flood pixels (n = 96) over the 60-day window centred on the cyclone landfall date. Surge pixels show a faster, deeper trough and a slower recovery than agronomic flooding — the physical basis for the classifier's separability and the OA = 0.844 SAR-only robustness variant.

Description. *Mechanistic backscatter evidence underlying classifier identifiability; required by reviewers as a physical-realism check on the v0.3.0 random-forest model.*

Abbreviations. SAR = synthetic aperture radar; σ^0 = normalised radar backscatter coefficient (linear power form, displayed in dB); dB = decibels (logarithmic backscatter unit, 10 · log10[σ^0]); VV = vertical-transmit, vertical-receive co-polarisation channel; VH = vertical-transmit, horizontal-receive cross-polarisation channel; CR = cross-polarisation ratio (VH − VV in dB); ΔVH = post-onset minus pre-onset σ^0_VH; n = number of hand-labelled pixels per class (here n = 96); days = calendar days (time-series x-axis unit, ±30 d window around landfall); OA = overall accuracy (fraction of correctly classified test pixels); v0.3.0 = released version of the Module 02 random-forest classifier.